

Real-time monitoring of protein precipitation in a tubular reactor for continuous bioprocessing



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ABSTRACT

Real time monitoring is critical to apply continuous bioprocesses successfully. This requires the integration of flow cells into continuous reactors. Potential continuous steps include protein precipitation, used in plasma purification, antibody enrichment and virus inactivation. Consequently we developed real-time monitoring of protein precipitation in a tubular reactor by pH, conductivity, turbidity and image processing. Feasibility studies were then performed for each monitoring method. Since integrated flow cells can affect time to reach steady state due to its inherent residence time distribution (RTD) property, we first determined flow behavior in flow cell is best described by laminar flow in pipes model, in order to find acceptable levels of volumetric flowrates that is not impractically high. Separately, we found pH highly important for protein precipitation while conductivity has high sensitivity to changes in buffer content. Turbidity signals are lumped values of floc properties while image processing can separately measure flocs by area, size and count. Also, the time and behavior to reach steady state amid changing precipitating conditions was investigated, to select for correct collection times to ensure high product quality. Lastly, we report the technical considerations in reactor design, construction and operation.

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1. Introduction

The benefits of continuous processing of biopharmaceuticals include universal platform, steady state, closed systems, easier scale up, minimized hold times, continuous flow, high volumetric productivity, flexible capacity, simplified operation and is compatible with disposable technology [1–3]. Examples include refolding [4,5], precipitation [6,7] and chromatography [8–10]. Economically, continuous processing can reduce cost of goods in different process scenarios [11]. Furthermore, continuous processing for biosimilars becomes economically favorable as the biopharmaceutical sector matures [12].

Among the bioprocessing steps, protein precipitation is simple and versatile. It can enrich proteins [13], remove contaminants [14], and allow selective precipitation of proteins [15]. Economically, it is an alternative to expensive chromatographic resins [16], where precipitation could replace the initial chromatography steps (e.g. ion exchange for capture). Furthermore, the mixing of protein solu-

tion to precipitant to desired conditions in precipitation enables easy transfer from batch to continuous reactors [6,7,17].

However, industrial application of continuous processes can only be realized when quality by design (QbD) requirements are met [18,19]. Particularly, continuous bioprocessing in downstream require better process analytical technology (PAT), with reliable control and hardware than is currently available [12]. This requires real time monitoring of process, where critical process parameters can be automatically adjusted in response to disturbances to ensure that the quality attributes consistently conform to the established acceptance criteria, required for industry to adopt continuous process [20].

Consequently, as an extension to previous work [5,6], we integrated several inline monitoring probes and used image processing for monitoring the precipitation step in a tubular reactor. Feasibility studies were then performed for each monitoring method. Specifically, we investigated 1) flow behavior of processing liquid caused by integrated flowcells, 2) characterize the different monitoring methods based on pH, conductivity, turbidity and image processing, 3) time to reach steady state amid changes in process conditions and 4) relationship between process conditions (pH, protein concentration) to precipitate properties via turbidity measurements and image processing. Additionally, we report the technical considerations for future work to better satisfy QbD and PAT paradigm.

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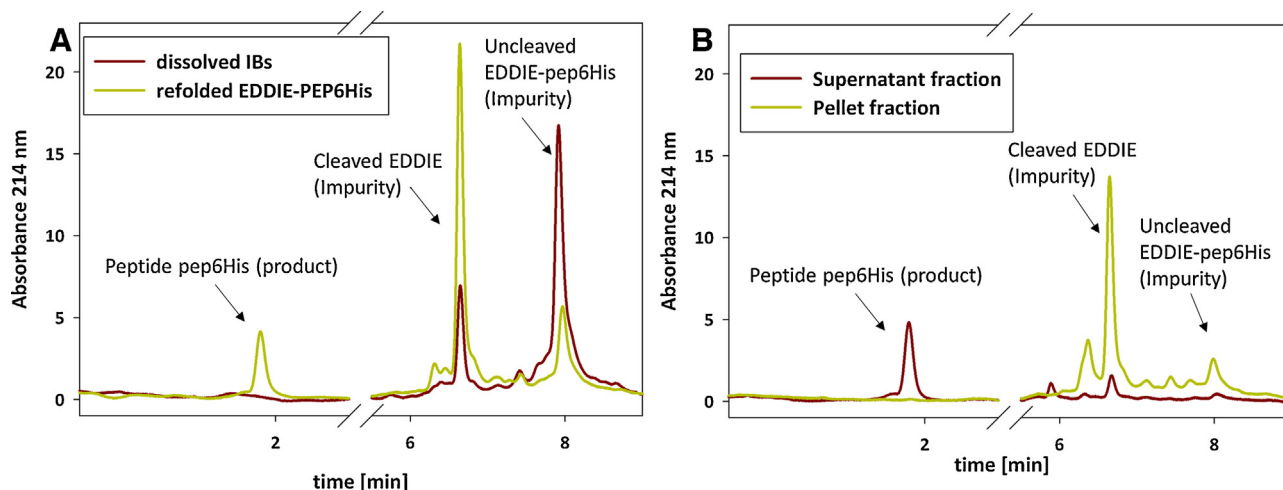


Fig. 1. Representative chromatograms; A) for dissolved IBs before refolding and refolded EDDIE-pep6His. B) Targeted peptide (pep6His) is purified and remains soluble in supernatant after acid precipitation at pH 4.4.

In current study, precipitation was performed in a small scale tubular reactor. The model protein used to perform precipitation was the refolded autoproteolytic fusion protein EDDIE-pep6His, which was overexpressed as inclusion bodies (IB) in *Escherichia coli* [21]. Under denaturing conditions of the dissolved IBs, EDDIE remains fused to its peptide partner, pep6His (a short 16 amino artificial peptide consisting of 10 randomly chosen amino acids and a C-terminal polyhistidine tag) [21]. However under refolding conditions, EDDIE releases its fused target pep6His (Fig. 1A). After refolding, the cleaved EDDIE and remaining uncleaved EDDIE-pep6His was precipitated out by lowering processing solution to below pH 5.0, while pep6His product remains soluble in supernatant (Fig. 1B). In this study, the precipitation step was performed in a tubular reactor and monitored in real time.

2. Materials and methods

2.1. Chemicals

All chemicals were purchased from Merck (Darmstadt, Germany) and Sigma (Steinheim, Germany) respectively, if not indicated otherwise.

2.2. Recombinant protein expression and IBs isolation

The recombinant protein EDDIE-pep6His were overexpressed in *E. coli* BL 21 with a pET30a plasmid (Novagen, Madison, WI, USA) containing the corresponding coding gene [21]. *E. coli* fed-batch cultivation was performed with a semi-synthetic medium on a 5-L scale according to Kaar et al., and isolation of inclusion bodies (IBs) was performed with an APV 2000 lab homogenizer (Invensys, Albertslund, Denmark) [22].

2.3. Batch dissolution and refolding of EDDIE-pep6His

Refolded material of EDDIE-pep6His for precipitation in tubular reactor was prepared by dissolving 1 part IBs to 4 parts dissolution buffer containing 10 M urea, 50 mM Tris and 100 mM monothioglycerol (MTG) at pH 7.3. To measure protein concentration, dissolved samples were filtered and measured on a Cary 50 UV–vis Spectrophotometer using the theoretical extinction coefficient $1.098 \text{ (mg/mL protein)} \text{ cm}^{-1}$ at 280 nm. Subsequently, overnight refolding was initiated by mixing 1 part dissolved pro-

tein to 6 parts refolding buffer containing 1 M Tris, 0.25 M sucrose (Acros Organics), 2 mM EDTA, 20 mM MTG (pH 7.3).

2.4. Residence time distribution (RTD) analysis of flowcell

A tubular reactor was set up with one IPC-4 peristaltic pump (ISMATEC, Wertheim, Germany), short hose of Tygon tube at 3.2 mm inner diameter (ISMATEC, Wertheim, Germany), respective flowcells shown in Fig. 2 (Swagelok W/TWO TUB (Hamilton) and self-designed turbidity flowcell (Bilek and Schüll GmbH, Vienna, Austria) and a fraction collector (Frac-920, GE Healthcare) (Fig. 3). To evaluate the residence time and the effective volume of the flow-cell positive step input experiments at two different flow rates, 1 mL/min and 5 mL/min were performed. A distinct amount of Vitamin B12 dissolved in 1x PBS buffer served as positive step input. The whole set up was filled with 1x PBS buffer before the Vitamin B12 input was started. The solution was pumped through the flow-cell at the respective flow rate and collected in fractions (a 0.6 mL and 0.5 mL) after exiting the flowcell by mean of a fractionizing collector. Absorbance at 550 nm of each fraction was measured by NanoDrop ND-1000 Spectrophotometer (Thermo Fisher Scientific) and concentration calculated using a self-established calibration curve. Concentrations were normalized by the initial concentration of the step input and plotted against time in order to obtain the cumulative distribution function $F(t)$. Subsequently, the best fitting model for the $F(t)$ curve was investigated using excel solver and table curve 2D (Systat Software).

2.5. Construction and operating parameters of tubular reactor

Using data from RTD analysis, acid precipitation experiments were carried out in the tubular reactor at a total flow rate of 5.00 mL/min. Tubular reactor was set up using IPC 4 and IPC-N peristaltic pumps (ISMATEC, Wertheim, Germany), GXM mixer (Stamixco, Winterthur, Switzerland), Tygon tubes of 3.2 mm inner diameter (ISMATEC, Wertheim, Germany), Swagelok W/TWO TUB flowcell equipped with pH and conductivity sensor (Hamilton, Bonaduz, Switzerland) and fractionizing collector (Frac-920, GE Healthcare). Conductivity and pH in the flowcell were recorded by Hamilton Device Manager (HDM) software every 5 s. Conductivity sensor was calibrated offline and temperature compensation factor was set at 2%/°C.

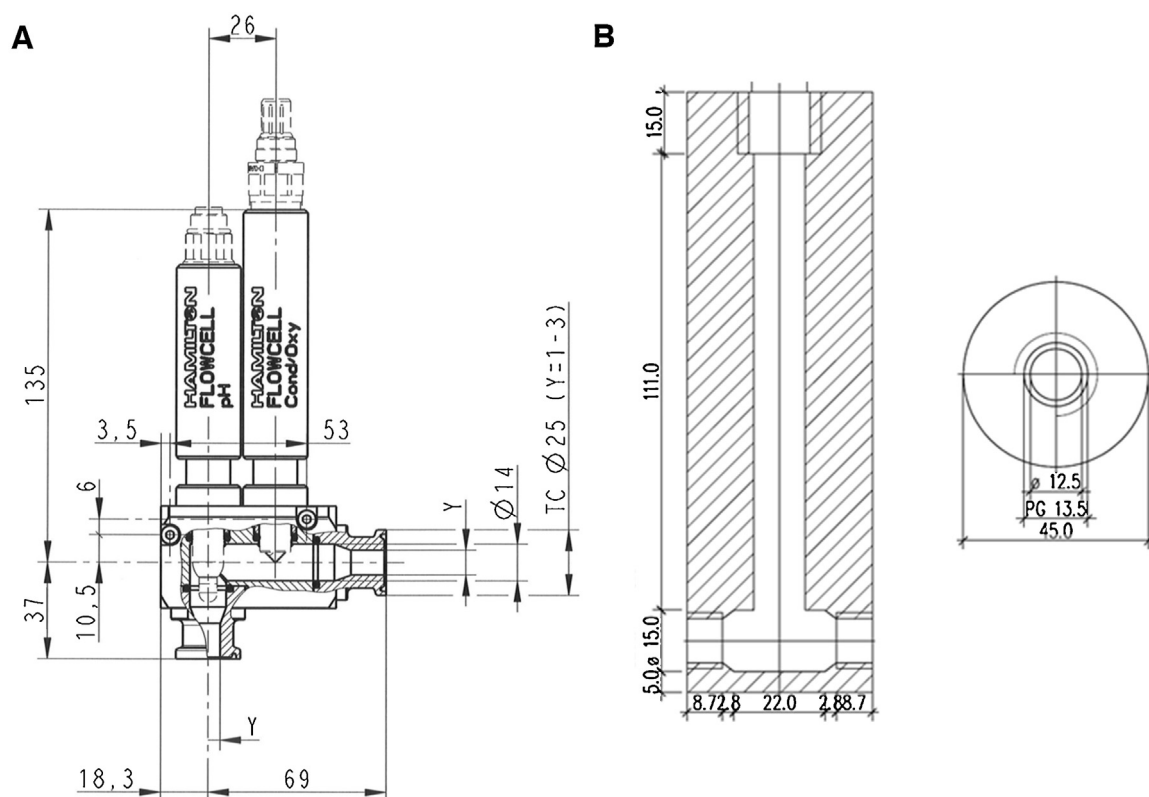


Fig. 2. Cross section and dimensions of A) Swagelok W/TWO TUB and B) self-designed turbidity flowcells in a tubular reactor.

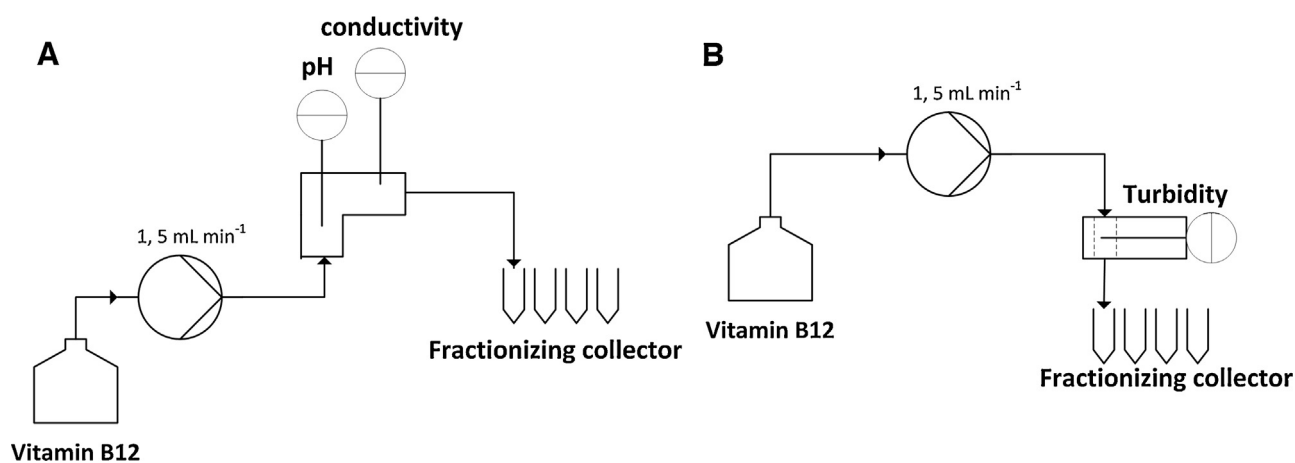


Fig. 3. Tubular reactor set up of step input experiment of vitamin B12 solution to determine the residence time distribution (RTD) of A) Swagelok W/TWO TUB and B) self-designed turbidity flowcells in a tubular reactor.

2.6. Titration to determine acid and protein ratio to attain desired pH

Operation parameters for the acid precipitation in tubular reactor were set on basis of a performed titration. Titration was performed where 200 mL buffer (V_{total}) made up of 171.4 mL refolding buffer and 28.6 mL dissolution buffer (seven-fold dilution) were stirred at 250 rpm while acetic acid was added in portions of 3 mL every 20 s. pH and conductivity were recorded in file by HDM software every 5 s. Temperature compensation factor was set at 2 %/°C. Subsequently respective flow rates for continuous acid precipitation were calculated on basis of the titration curve.

2.7. Analytics of supernatant and precipitate

Fractions at 30 mL process solution were collected at the end of the tubular reactor using a fractionizing collector and HPLC analyses of supernatant and dissolved pellet were carried out as described in Ref. [6]. Samples of supernatant for HPLC analysis were either filtered at 0.22 μ m or centrifuged at 4700g for 15 min. Pellet, dissolved in 30 mL dissolution buffer, and supernatant were analysed separately. Fractions of the precipitation experiment at pH 5.8 were centrifuged for additional 15 min at 4800g.

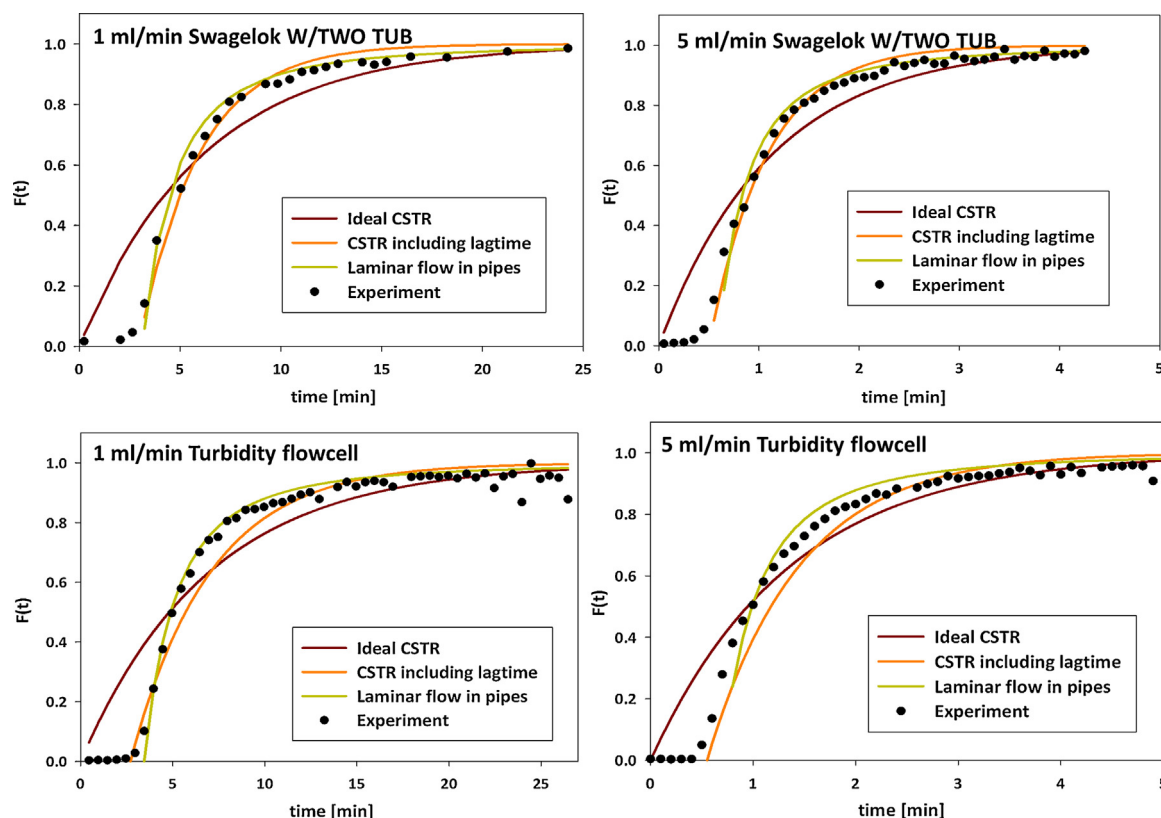


Fig. 4. Schematic diagram of protein precipitation of refolded EDDIE-pep6His using acetic acid in a tubular reactor, equipped with a A) static or B) active mixer at the initial stages of precipitation with inline and image processing equipment.

2.8. Turbidity monitoring on the continuous precipitation of EDDIE-pep6His

For turbidity monitoring, an additional self-designed turbidity flowcell fitted with InPro8050 turbidity sensor (Mettler Toledo, Urdorf, Switzerland) was integrated into the rear section of the tubular reactor (Fig. 4). Conductivity and pH monitoring was started as soon as the flowcell was filled with process solution. Measurements were recorded in file every 5 s. Inline calibration of conductivity sensor was activated and temperature compensation factor was set at 2%/°C. Turbidity monitoring started once the flowcell was filled (approx. 26.5 min after pH monitoring started). Times through different sections of the reactor (e.g., from junction to flowcell) were measured by means of a stop watch and used for subsequent data evaluation. Thus, all process events (change of flow rates) and recorded turbidity and conductivity signals were related to the pH monitoring time.

Specifically, EDDIE-pep6His precipitation using 1 M acetic acid was performed at pH 4.2 and pH 5 for two different protein concentrations (0.734 and 0.367 mg/mL) (in total four process runs). Acetic acid and refolded protein were pumped at these flow rates through the tubular reactor for defined time intervals. The total volumetric flow rate was kept at 5 mL/min over the whole process.

Turbidity sensor was calibrated inline by means of a manual two point calibration as suggested by the technical support of Mettler Toledo. Zero point (=offset) was set as water was pumped through the flowcell at 5 mL min⁻¹. The slope and thus 100% were defined by pumping precipitated EDDIE-pep6His (0.751 mg mL⁻¹, precipitated in beaker at pH 4.2 for 30 min) through the cell at 5 mL min⁻¹ and setting the slope (=21,700).

2.9. Image processing setup

Image processing setup was similar to previous work. Precipitate growth at pH 4.4 in tubular reactor was captured within glass rods (3.2 mm i.d.) at residence times 3.3, 8.3 and 28 min that are connected to 3.2 mm Tygon tubings. These glass rods were laid beside each other and pictures taken from above with a mounted macro camera (Olympus OM-D 50 mm 1:2 MACRO ED lens) at steady state during the precipitation process. ISO was set at 200, aperture between f/11-f/14, shutter time at 1/250 s. A cover is placed over the camera flash and glass tubes to minimize external light source. Flash (FL-50) was angled such that there was enough light but no direct reflection from glass tubes into camera lens. A black surface was placed as a background from the tubes to allow better contrast from the white precipitates.

Images were analysed with ImageJ. These were converted to 8 bit; background was then subtracted, converted to binary before results within tubes were calculated. Many images can be processed automatically from recorded macros within a folder. This ensures image processing are consistent and can be done at high throughput. In each processed binary image, the particle count, normalized area covered by particle and average particle size is being analysed. In each analysis of a specific condition, the average of ten images was used, which were taken over 30 min when steady state was achieved.

3. Results and discussion

The key objective was to develop and investigate monitoring systems for protein precipitation via inline probes (pH, conductivity and turbidity) and image processing. Due to increased residence time distribution of process fluids within integrated flowcells in continuous reactors, we investigated fluid flow of two flowcells

Table 1Summary of results of the RTD analysis obtained for the Hamilton flow cell by fitting different $F(t)$ models.

Flow rate [mL/min]	$F(t)$ model [–]	Space time τ [min]	$V_{\text{effective}}$ [mL]	Sum of squares [–]	Lagtime t_L [min]
1	CSTR	6.1	6.1	0.314	–
	CSTR-lagtime	5.9	5.9	0.052	2.9
	Laminar flow pipe	6.3	6.3	0.027	–
5	CSTR	1.1	5.6	0.346	–
	CSTR-lagtime	1.1	5.4	0.044	0.5
	Laminar flow pipe	1.2	5.9	0.037	–

via residence time distribution analysis. After, we established the relationship between pH monitoring and product purity during the continuous precipitation process of refolded EDDIE-pep6His (Fig. 1). Subsequently, we determined how fast steady state can be attained amid changing conditions. Lastly we characterize data from turbidity and image processing to precipitate properties.

3.1. Continuous precipitation in a tubular reactor setup

We first define the process conditions of our continuous precipitation step. The aim of our acid precipitation step is to remove cleaved and uncleaved EDDIE (impurities) by acid precipitation while keeping pep6His (product) soluble in solution (Fig. 1). Previous study showed correct pH levels will precipitate out impurities while keeping product in solution [6]. Over a process run, precipitation performance is determined offline by HPLC analysis, detecting the presence of product and impurities, while pH was monitored inline by pH sensor, fitted into flow cell. Additional parameters based on conductivity, turbidity and image processing were investigated.

3.2. Residence time distribution (RTD) analysis of flowcells

Integration of inline monitoring systems into the tubular reactor will exacerbate RTD of process fluid. Therefore, it is beneficial to determine a feasible RTD at a sensible flowrate of a flowcell. Depending on the flow pattern and reactor type, the RTD for a fluid element differs [23]. Since this is an issue that affects all inline monitoring systems, we determined experimentally the best fitted model to describe the RTD behavior and fluid flowrate.

The RTD in this study is used to predict changing pH conditions, which are instantaneous when the acetic acid meets the buffer solution containing the dissolved protein. The flow rate distribution can thus determine how fast the desired pH condition is reached, which in turn affects precipitation performance. The fitted models are limited to instantaneous reactions like pH change. Precipitate growth prediction, beyond the scope of this study, would require integration of precipitation kinetics into RTD equations. Specifically, step input experiments—switching from phosphate-buffered saline (PBS) to PBS containing vitamin B12 buffers at 1 and 5 mL/min flow rates, were performed to obtain the cumulative distribution $F(t)$ of vitamin B12 in Swagelok W/TWO TUB (Hamilton) and self-designed turbidity flowcells (Fig. 3).

$F(t)$ is dimensionless, describing the cumulated fraction of vitamin B12 leaving the reactor within a certain time. For example, when $F(t)$ is 0.9, the outgoing vitamin B12 concentration at a specific flow rate and residence time is 90% of the stock solution. Experimentally, this was determined from the percentage of vitamin B12 collected in fractions over time. Subsequently, we fitted Eqs. (1)–(3) describing different reactor models to determine the best flow behavior in the flowcells:

Mixed flow in ideal CSTR:

$$F(t) = 1 - \exp\left(-\frac{t}{\tau}\right) \quad (1)$$

CSTR including lag time t_L to account for minimum time liquid travels through flowcell:

$$F(t) = 1 - \exp\left(-\frac{t + t_L}{\tau - t_L}\right) \quad \text{for } t \geq t_L \quad (2)$$

Laminar flow in pipes:

$$F(t) = 1 - \frac{\tau^2}{4 \times t^2} \quad \text{for } \frac{t}{\tau} \geq 0.5 \quad (3)$$

In Eqs. (1)–(3), $F(t)$ is the cumulative distribution, t [s] is time and τ [s^{−1}] is space time (time necessary to process one flow-cell volume). Fitted data showed that laminar flow in pipes best describes the fluid flow in both flowcells (Fig. 5). This indicates that the flowcell behaves predominately as a pipe itself and not as a CSTR. Similarly, the sum of squares between the respective models and experimental results were lowest in the laminar flow in pipes model (Table 1). Concurrently, Eq. (4) calculates the effective volume of the flowcell:

$$V_{\text{effective}} = \tau \times \dot{v} \quad (4)$$

where $V_{\text{effective}}$ [mL] is the effective volume in flowcell and $\dot{v}$ [mL/min] is the volumetric fluid in flowcell. Since the best fitting $F(t)$ model for a flowcell is best described as “laminar flow model in pipes”, the cumulative distribution at different flow rates and residence time in flowcell could thus be estimated. Rearranging Eqs. (3) and (4) gives:

$$t = \frac{0.5 \times \sqrt{\frac{1}{1 - F(t)}}}{\dot{v}} \times V_{\text{effective}} \quad \text{for } \frac{t}{\tau} > 0.5 \quad (5)$$

This can be graphical represented in Fig. 6, where residence times for $F(t) > 0.90$ are impractically high at volumetric flow rates below 4 mL/min. However, residence times decrease to an acceptable level above 4 mL/min. Further increase of flow rate showed a smaller reduction in residence time. Consequently, volumetric flow rates at 5 mL/min were used for subsequent operations. In this respect, residence times within the flowcell can be kept in an acceptable range without excessive consumption of process volumes.

3.3. Relationship of pH and conductivity monitoring to product quality

Previous work showed that pH 4–5 was vital in precipitating out impurities (cleaved and uncleaved EDDIE), while the target peptide pep6His remained soluble [15]. In this study, continuous precipitation in a tubular reactor was initiated via pumping acetic acid and

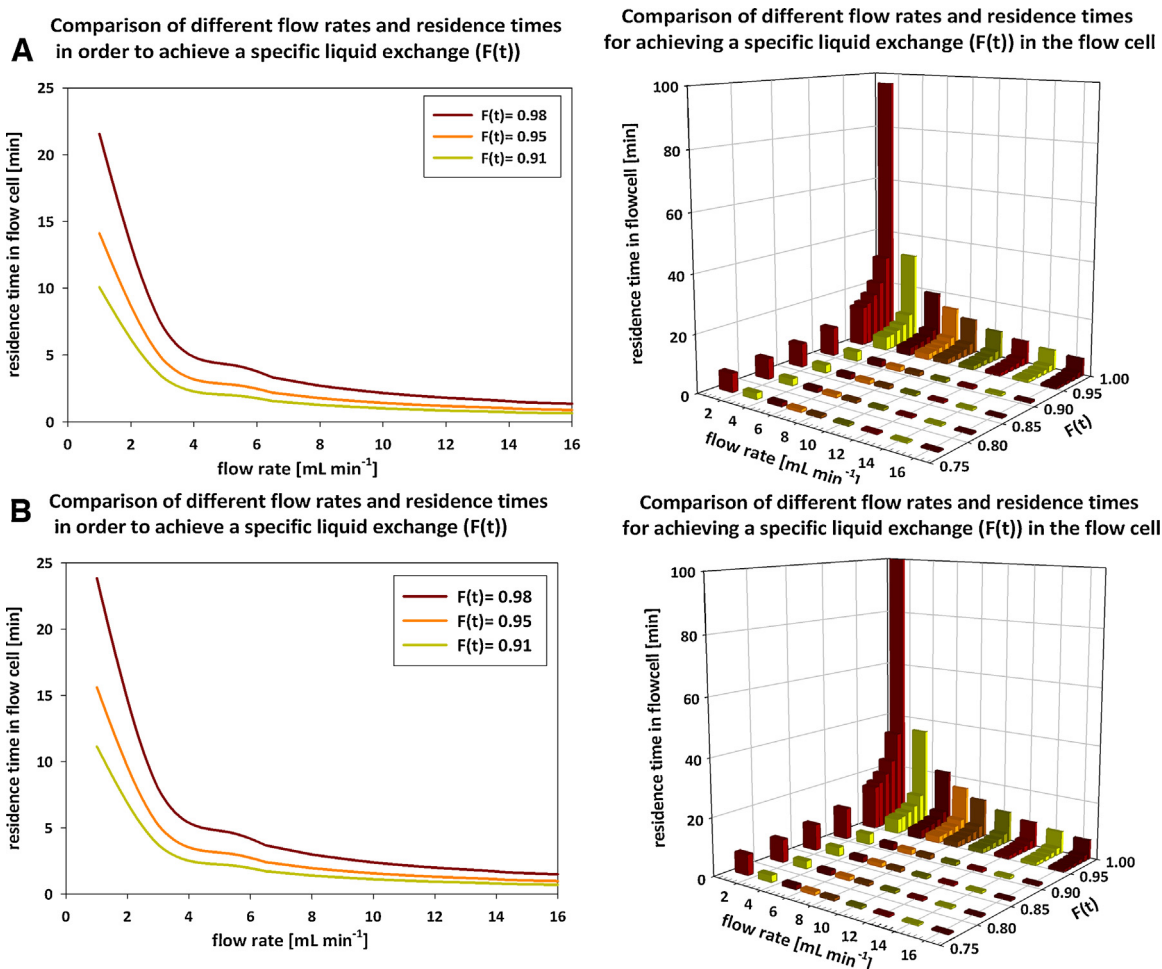


Fig. 5. Residence time distribution of experiment and different fitted models of Swagelok W/TWO TUB and self-designed turbidity flowcell in step input experiments at 1 and 5 mL/min.

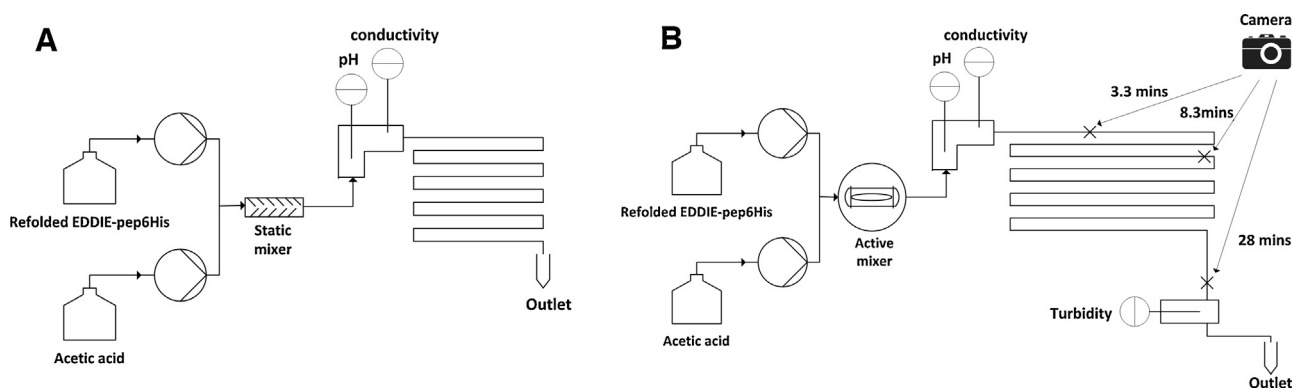


Fig. 6. Comparison of different flow rates and residence times for achieving specific liquid exchange $F(t)$ in A) Swagelok W/TWO TUB and B) Turbidity flowcell.

refolded EDDIE-pep6His via a GXM static mixer to obtain a final pH of 4.4 and 5.8. Pump flow was calculated on basis of a pre-defined titration curve. Solution was also monitored via a flowcell equipped with pH and conductivity probes (Fig. 4A). Concurrently, fractions of exiting solution were analyzed to determine precipitation performance.

Our product (peptide pep6His) remained 100% soluble in the supernatant for different precipitating conditions. However, the removal of impurities (cleaved EDDIE and uncleaved EDDIE-pep6His) was different in pH 4.4 and 5.8. At pH 4.4 impurities were

mostly precipitated out from supernatant. In contrast at pH 5.8, impurities did not precipitate and remained in solution (Fig. 7). In this regard, monitored pH could be correlated to precipitation yields and thus is a significant monitoring parameter for acid precipitation. Furthermore, pH measurements were robust over operation time (90 min). Additionally, accurate pump flow was essential to the mixing proportion between acetic acid and refolded solution, which affects pH and thus precipitation performance.

We also monitored conductivity in our process solution. Unlike pH which was more stable, measured conductivity increased

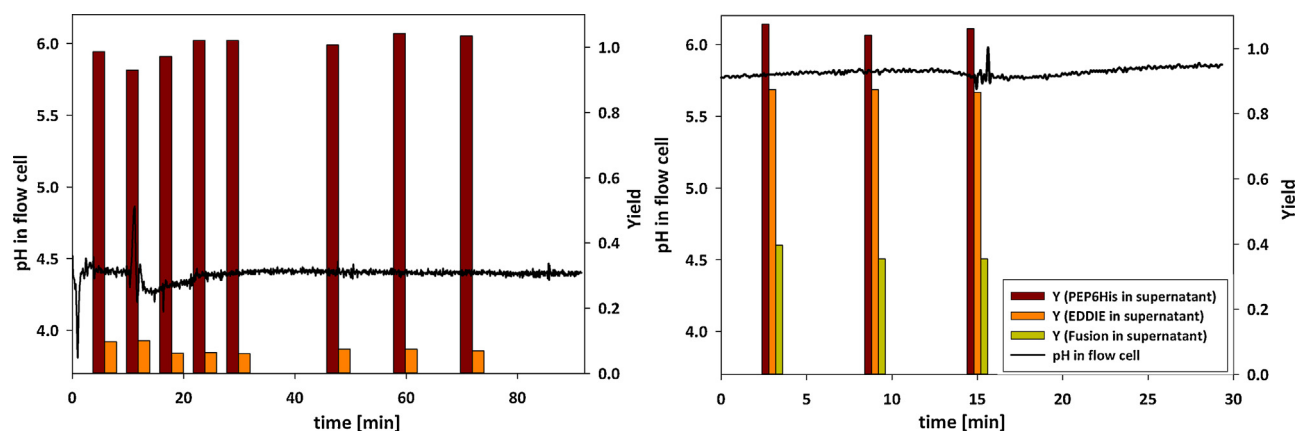


Fig. 7. Precipitation yields of EDDIE, pep6His and fusion protein in supernatant in relation to measured pH in the flowcell; left: acid precipitation at pH 4.4, right: acid precipitation at pH 5.8.

slightly over time (data not shown). Additionally, the conductivity sensor was not calibrated inline and thus only relative measurements could be performed. The reason for the steady signal shift was unclear, although insufficient mixing could be a reason. Small flocs were present in the initial feeding section of the tubular reactor, and some were trapped within the GXM mixer and accumulated over time. As a result, mixing efficiency of the static mixer would decrease. In this regard, GXM mixers are not suited for mixing process flows with solid particles in small scale. The decrease in mixing quality could account for the conductivity signal shift.

In summary, acid precipitation in tubular reactor with inline pH and conductivity monitoring system was performed. Precipitation yields were comparable to previous studies [6]. Inline pH measurements were robust over operation time and thus well suited for process monitoring. Conductivity drifting could be from insufficient mixing and static mixers were replaced with active mixers in subsequent studies.

3.4. Attaining steady state amid changing process conditions

Ideally, continuous process would operate at true steady-state, keeping operating conditions and processing solution constant [24]. Therefore it is important to only collect processing solution where steady state is established. To understand how fast steady state can be attained, we switched on and off the acid feed at distinct time intervals during precipitation process. This was performed with pH, conductivity sensors and an additional turbidity monitoring in the exiting section of the tubular reactor (Fig. 4B). Inline calibration for conductivity sensor was conducted for more reliable monitoring. To avoid precipitate from clogging in static mixer, an active mixer was used. Precipitation was performed at four conditions: pH 4.2 at 0.38 or 0.74 mg/mL and pH 5 at 0.38 or 0.74 mg/mL.

Inline monitored pH and conductivity responded to changes during all four acid precipitation runs (Fig. 8). The measured pH in the flowcell responded to changes in process conditions, similar to the expected pH investigated by means of titration. This was robust over the entire operation and thus suitable for monitoring continuous acid precipitation processes. However in most cases, a drop from a higher to lower pH resulted in a slower than expected response, where steady-state was established over 20 min. This slow response is unlikely due to the inherent response time to the pH probe, which is faster by two orders of magnitude. Instead, due to the RTD of the flow cell, a small portion of buffer solution at higher pH 7.3 still remained in the flow cell. Due to the high Tris content (1 M), a slightly higher amount of buffer solution is sufficient to keep the pH higher than expected. Therefore a much longer time was required to completely remove the additional buffer to attain

steady state. This example shows how RTD can delay steady state in continuous processes.

Almost the same trend could be seen for conductivity measurements. However, slight deviation (not more than 2 mS cm^{-1}) between expected and measured conductivity were observed and as acid flow was switched off, the conductivity signal did not reach initial and constant value again (Fig. 8). A steady conductivity increase was observed although acid supply was switched off. This was due to the lack of a valve to completely seal acid from the refolded protein where both fluids meet before the active mixer (Fig. 4B). Refolded protein flowing through the T-mixer would come into contact with the stationary acetic acid at the T-junction, affecting the conductivity of the solution. In contrast, pH did not decrease because of the high buffering capacity at pH 7.3 of 1 M Tris (pK_a of 8.1 at 25°C) of the refolding buffer. In this case, conductivity sensor was the more sensitive monitoring tool for detecting minute buffer changes in process solution than pH sensor.

In essence, the time to attain steady state should be found, so that the ideal time to collect exiting product can be determined. In particular, if continuous process deviates from process design space [18], process can immediately divert exiting solution away from the main collection pool. Once operating range is re-established, process can redirect solution back to collection pool. Such a strategy can increase the flexibility and robustness of a continuous process, without discarding the whole collection due to contamination, or restarting the continuous process. Longitudinal diffusion of solutes in solution in the tubular reactor should also be accounted for. This would disqualify a larger proportion of process solution when a deviation occurs, but will further increase the robustness of process.

3.5. Relationship between turbidity monitoring and precipitation conditions

Response time and feasibility studies on the turbidity monitoring system were investigated together in this study. Turbidity sensor is based on backscatter technology measuring particle size and count using near infrared light at 880 nm. The flowcell suitable for inline monitoring at laboratory scale was self-designed. In contrast to pH and conductivity monitoring, turbidity was measured in the rear section of the tubular reactor (after residence time of approximate 30 min) where precipitation have reached equilibrium [6]. Differences in turbidity measurement reflected the different process conditions on pH and protein concentration (Fig. 9).

Precipitation at pH 4.2 and 0.367 mg mL^{-1} was first performed (Fig. 9A) where the flowcell for the turbidity sensor was kept in inclined position (approximate 30° inclination) for the first

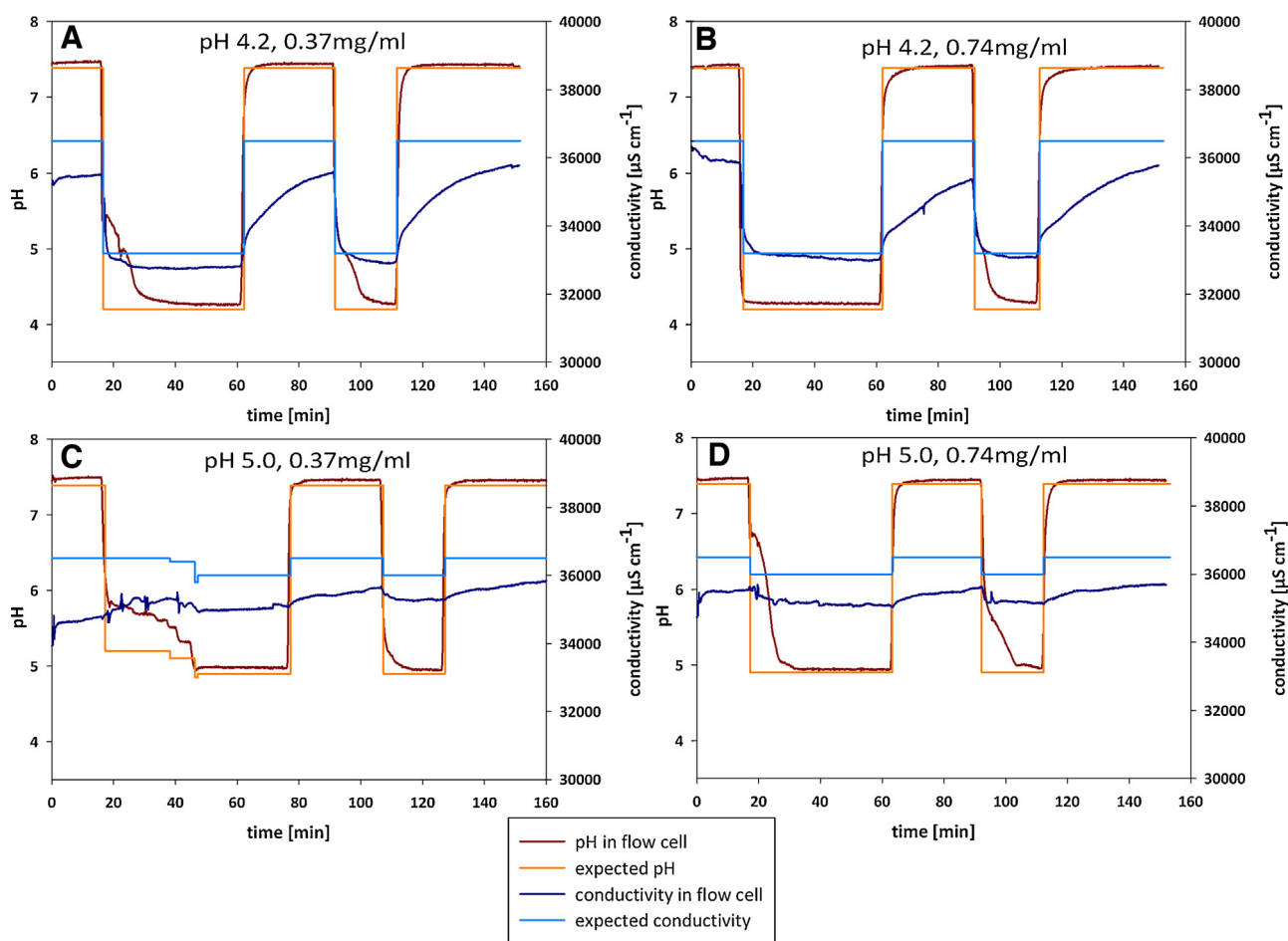


Fig. 8. pH and conductivity measurements in the flowcell during continuous EDDIE-pep6His precipitation at different operation conditions.

105 min. A noticeable rise in turbidity signal was observed between 20–60 min. Furthermore, turbidity signals resulted in a series of peaks between 60 and 90 min even though no precipitates were visible in the process solution (as acid was switched off). This signal profile was the result of accumulated precipitates within the flowcell and thus no reliable measurements could be obtained. To remove accumulated precipitates, flowcell was positioned vertically so that the flow path was in vertical downwards direction. Consequently, turbidity signal decreased to an almost constant value close to zero as no precipitates showed up for the last 15 min of operation. In this regard, it was shown that the position of the flowcell has a big impact on turbidity measurements. Only by keeping the flowcell in vertical position, adequate turbidity measurements can be made. This was the case for the other three process runs at pH 4.2 (0.736 mg mL^{-1}) and pH 5 (0.367 mg mL^{-1} , 0.736 mg mL^{-1}) (Fig. 9B–D).

The more precipitates present in solution the higher was the averaged monitored signal. Monitored signals however, resulted in a series of peaks, which showed variations in their height over operation time. Only a semi-quantitative correlation between signal and amount of precipitates could thus be inferred. More data might be needed in order to evaluate whether a quantitative statement can be deduced or not. As no precipitates were present, turbidity signal reached a constant value close to zero. Furthermore, the turbidity sensor was calibrated relatively as suggested by manufacturer. Thus, the measured signal was only a relative value and cannot be referred to a certain precipitate amount.

Altogether, turbidity monitoring of precipitation processes was suitable to monitor whether precipitation occurs or not; turbidity

monitoring using near infrared light at 880 nm does not provide quantitative resolution. However it still complements pH and conductivity monitoring resulting in a more powerful monitoring system for pH dependent precipitation reactions. Furthermore, a more reproducible calibration method or rather calibration with appropriate standards might be needed. Secondly, turbidity measurements alone do not represent a robust monitoring system. Signals were highly fluctuating and variations in flow rates might have an impact on measured signals. In addition to that, size and structure of precipitates influence turbidity readings next to protein concentration [25]. Since signals were lumped process parameters (concentration, pH, flow pattern), a significant data interpretation might be hard to achieve.

3.6. Characterization of precipitates via image processing

We defined image processing as images taken with a mounted camera and the subsequent image processing with appropriate software. This allows monitoring without direct contact, thus eliminating any contamination risk. We investigated the feasibility of image processing in continuous precipitation where the particle count, particle size and area covered by particles was monitored. This was performed at pH 4.0 at 0.15, 0.29, 0.58 mg/mL protein concentration and pH 5.6 at 0.58 mg/mL (Fig. 10A), using precipitation conditions from previous studies [6].

The main purpose was to explore the potentials of precipitate monitoring using image processing methods. Although a system can only give an apparent value for each parameter, such a correlation between precipitating conditions could be established.

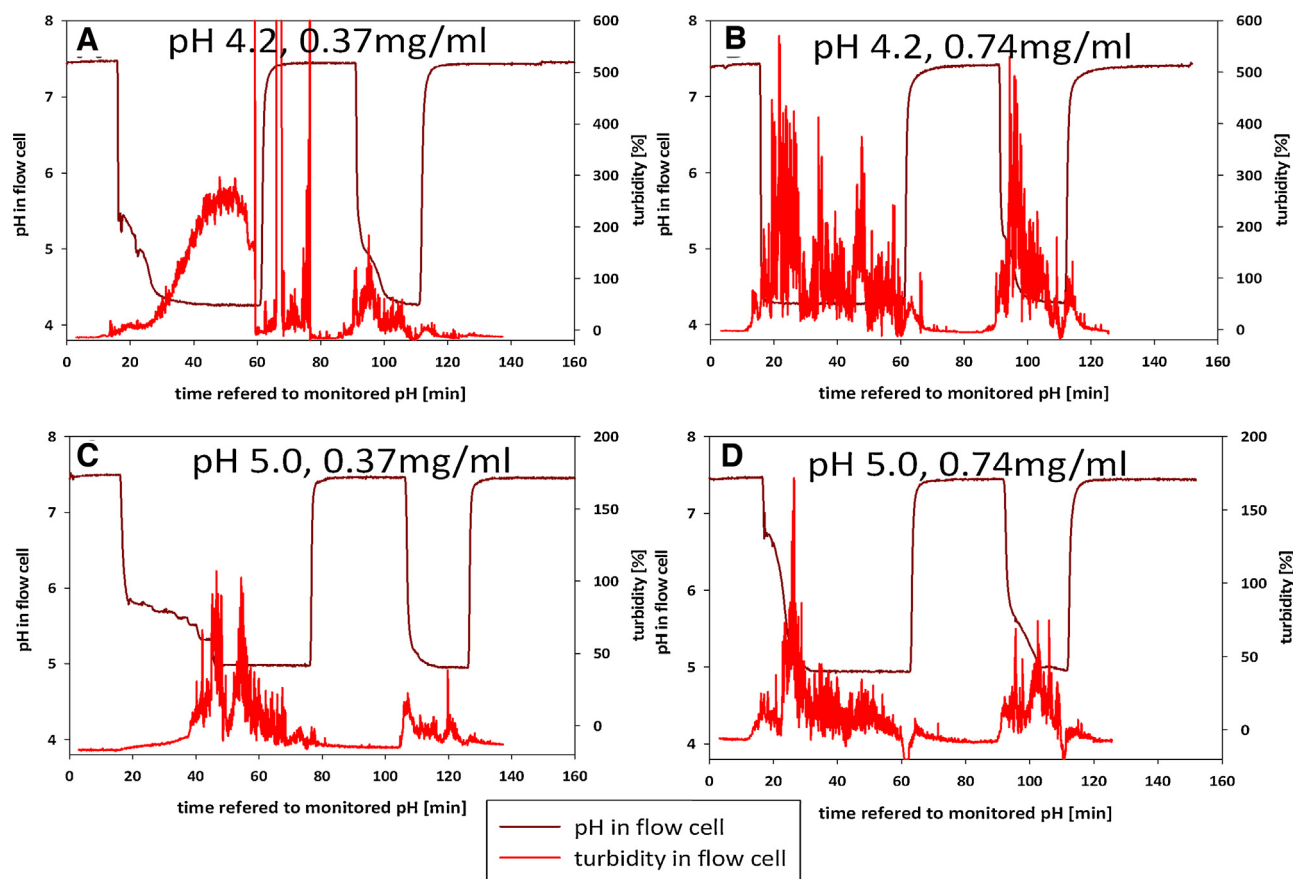


Fig. 9. Turbidity measurements in the flowcell during continuous EDDIE-pep6His precipitation at different operation conditions in relation to monitored pH. A) Position of flowcell for turbidity sensor was changed from vertical to horizontal at 80mins process run at pH 4.2, 0.37 mg/mL protein precipitation, to avoid accumulation of precipitates in the flowcell. This position was kept for runs B), C) and D). Note: different scaling of turbidity axis at pH 5 and pH 4.2.

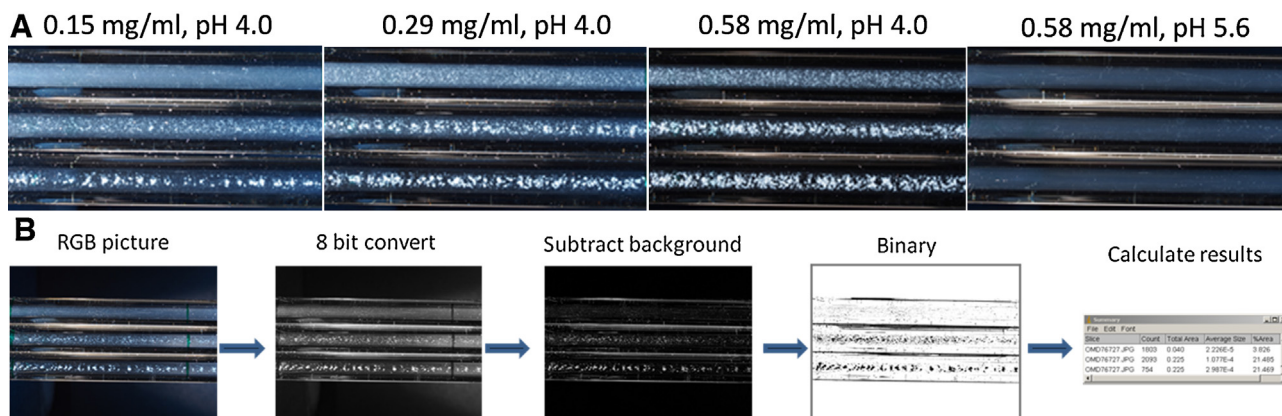


Fig. 10. A) Images of precipitates at different residence times at different protein and pH conditions in a tubular reactor. Each tube image captured was 4.7 cm long. The top, middle and bottom tubes represents 3.3, 8.3 and 28 min residence times respectively. B) Images were analysed with ImageJ. These were converted to 8 bit; background was then subtracted, converted to binary before results within tubes were calculated. In each processed binary image, particle count, normalized area covered by particle and average particle size were analysed.

Specifically, an image processing monitoring system was set up (Fig. 4B), experiments were performed and precipitating conditions were verified with pH and conductivity sensors. Once steady state was reached, images were subsequently analyzed to observe the trends of particle characteristics at different precipitation conditions and protein concentrations (Fig. 10B).

At pH 4.0, residence times taken in each image were approximately 3.3, 8.3 and 28 min. Since particles were observed at 3.3 min, precipitation growth governed by fluid motion was already taking

place. This happens after particles have reached a critical size, typically 1 μm in diameter. In this growth regime, particles tend to grow by colliding and then sticking together. This is a precipitation process where electrostatic repulsion between particles is reduced in comparison to the attractive van der Waals force [26,27]. The general observations showed that over precipitation time, number of particles decreases, while normalized area covered by particle increases and particle size increases. Therefore the calculations

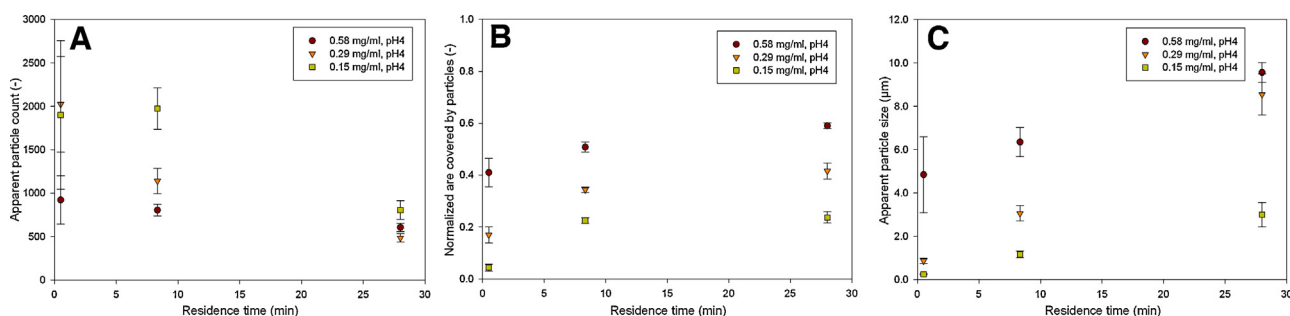


Fig. 11. Image processing experimental data in precipitation conditions of 0.58, 0.29 and 0.15 mg/mL protein concentration at pH 4 by: A) particle count, B) normalized area covered by particles and C) particle size. Ten images were taken over 30 min when steady state was achieved. Each data point represents the average of ten images of one standard deviation.

between the analyses were in line with each other, and in agreement with the precipitation mechanism.

In comparison at different protein concentrations in the precipitation process, higher concentrations showed less particle count (Fig. 11A), particles cover more normalized area (Fig. 11B) and average particle size was larger at each residence time (Fig. 11C). Visual observation at pH 5.6 showed particle count, normalized area and size to be negligible (Fig. 10A). In this case, imaging software cannot quantify images below its limit of detection. Overall, we showed that image processing can characterise precipitates into three separate parameters, in contrast to a lumped parameter by turbidity monitoring.

Therefore, the quality of precipitates can be confirmed by image processing when all three parameters described meet predefined values of a specific process. Using this monitoring method, future process control could be implemented when the amount of acid, which affects pH and thus precipitation, is automatically adjusted in response to any deviation monitored via image processing, thus ensuring that precipitate quality is consistent during continuous process.

Other than precipitation, image processing could be applied as additional quality control in other bioprocessing steps such as protein dissolution [28], protein refolding [29,30] and protein crystallization [31].

3.7. Technical considerations for monitoring continuous precipitation in a tubular reactor

Air trapped in flowcell could affect inline measurements because sensors will measure the empty space within the flowcell and not the process fluid. At the start of the process, flowcell should be fully flushed with process solution. Any trapped bubble can be released by tilting the flowcell upwards for bubble to flow out. Alternatively, a bubble trap device could be integrated right before the flowcell.

Similarly, precipitate trapped in the flowcell can affect inline measurements. This was a concern for the turbidity flowcell that

was placed at the rear end of the tubular reactor where precipitates were larger than at the start of the tubular reactor. Positioning the turbidity flowcell such that the flow path is in a vertical downward direction resolves any accumulation issues.

Active mixer was a more effective mixer than the GXM static mixer in continuous acid precipitation. On basis of measured pH and conductivity and comparison to values from titration, the active mixer seemed more capable of mixing process solutions efficiently. Furthermore, the active mixer demonstrated to be an appropriate mixing device in small scale for blending solutions with solid particles. No precipitates got trapped within the mixer and thus previously described problem with GXM mixer could be solved.

Monitoring accuracy was improved when sensors were calibrated inline because this mimics the monitoring environment of the flowcell. Leaking of flowcells can be prevented by using the correct 'O' rings to ensure proper sealing. Vaseline and thread seal tape would allow temporary sealing from unforeseen leakages.

During start of precipitation reaction, an accumulation of precipitates was formed at the front section of the fluid. Larger precipitates protrude into the higher velocity region of laminar flow, hence were carried further in a specific amount of time than their smaller counterparts. However, steady state should be attained after this accumulated region exits from reactor. Pumps capable to withstand increased back pressure caused by the accumulated precipitates should be used.

Keeping the residence time of the tubular reactor is crucial for product quality from bench to production scale. Additional residence time for inline flowcells must also be accounted for. Once this criteria is met, there are three ways of scaling the tubular reactor which are scaling up, scaling out and scaling linearly. These three methods are described and their advantages and disadvantages are listed in Table 2.

We performed precipitation using acetic acid as the precipitant. Performance may differ when using more viscous precipitate such as PEG. Higher viscosity could result in fluid moving slower than settings from the pumps, resulting in high back pressure. Lower viscosity precipitant could alleviate this issue. Active mixers are

Table 2
Methods to scale the tubular reactor.

Scaling method	Scale up parameter	Advantages	Disadvantages
Scaling up	- Inner diameter - Flow rate	- No increase in floor space	- Increase in axial diffusion - Decrease surface tension - Yields may differ
Scaling out	- Number of tubular reactor	- Identical flow pattern - Identical yields to bench scale	- More pumps required - Proportional increase to floor space
Scaling linearly	- Tube length - Flow rate	- Easy extension to existing tubular reactor	- Higher shear forces due to increase flow rate - Yields may differ

recommended to quickly initiate precipitation step and at the same time, decrease the overall viscosity from the mixing with protein solution.

3.8. Future work in continuous precipitation

Currently, feasibility studies of real-time monitoring methods for continuous precipitation were performed. To fully satisfy QbD and PAT requirements [32,33], steps should include: 1) risk analysis to better identify parameters for process characterization, 2) the use of design of experiments (DOE) to define design space, 3) studies executed and results analyzed to determine the importance of the parameters as well as their role in establishing design space. The operating space, as defined by the operating ranges, is then well nested inside the design space indicating robustness of the process step [18]. Subsequently, design space can be converted into window of operation diagrams [34], for intuitive data interpretation [35].

In continuous precipitation, parameters of interest would include pH, flow rate, protein concentration, acid concentration, conductivity, residence time and the dimensions of tube and flowcells. Furthermore, the frequency tests, robustness, reliability, accurate resolution, precision, sensitivity and response time of monitoring methods has to be statistically defined. Once real time monitoring is fully satisfied, feedback for automatic closed looped control can then be established [36]. In addition, other monitoring technologies can be implemented. Similar to real-time pooling of a liquid chromatography step [37], an online or at-line HPLC system could be used to analyze the exiting solution from the tubular reactor that are collected in fractions.

4. Conclusion

Feasibility studies of real time monitoring in continuous precipitation reaction in a tubular reactor was performed. A model to describe the residence time distribution of flowcells was defined and the performance of pH, conductivity, turbidity sensors and image processing were characterized. Time to achieve steady state amid changing conditions was investigated. Technical considerations to reactor setup and future work was further reported. The principals of this study could also be applied to other bioprocessing steps such as continuous protein refolding, protein dissolution and protein crystallization. We consider this study a contribution to develop real-time monitoring in continuous biomanufacturing.

Author contributions

M.Z. and S.P. were involved in experimental design, execution of experiments and modelling work. A.J. and R.H. conceptualized and supervised the work. All authors were responsible for preparation of the manuscript.

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